

APPROACH DOCUMENT: SHL ASSESSMENT RECOMMENDER BOT

PROBLEM & SOLUTION

Hiring managers struggle to find the right assessment from 377+ SHL tests without knowing exact keywords. I built a conversational bot that clarifies hiring needs and recommends real SHL assessments with direct links.

Architecture: FastAPI backend (/health and /chat endpoints) + Streamlit UI. Stateless: every request includes full conversation history.

DESIGN CHOICES & RETRIEVAL SETUP

Three components:

1. **FastAPI (api.py):** Handles endpoints, returns structured JSON (reply, recommendations, end_of_conversation)
2. **Bot Logic (bot_logic.py):** Filters 377-test catalog by keyword matching
3. **Streamlit UI (app_streamlit.py):** Chat interface for testing

Retrieval strategy: Load all 377 tests in-memory at startup. Detect tech keywords in user message (java, python, c#, etc.). Filter catalog by keyword and return top 10 with real names and SHL URLs. No external vector DB needed—simple string matching is fast enough.

Example: User says "java developer" → Detect "java" → Return 10 Java tests with links

BOT BEHAVIOR & PROMPT DESIGN

Decision logic:

1. Tech keyword detected? → Recommend tests immediately
2. First turn + vague (no keyword)? → Ask clarifying questions (role, seniority)
3. User says "done"? → End conversation
4. Default → Ask for more details

Example flow:

- User: "I need assessments for Python developers" → Bot detects "python" → Recommends 1 test immediately
- User: "I need a test" → Bot asks "What role? What seniority level?" (vague, first turn)

Scope: Only discusses SHL assessments. Refuses general hiring advice and prompt-injection attempts. Every URL comes from catalog (no hallucination).

CHALLENGES & SOLUTIONS

Problem	Cause	Solution	Result
Infinite loop on first message	<code>st.rerun()</code> re-sent message	Added message history check before processing	First response immediate, no loop
Bot asked questions even when tech specified	Only checked message length	Added keyword detection logic	Bot now recommends instantly for tech requests
JSON encoding error	Default encoding couldn't handle special chars	Used <code>encoding='utf-8'</code> when loading catalog	Catalog loads successfully

EVALUATION APPROACH

Testing methods:

- Swagger UI (localhost:8000/docs): Manual API endpoint testing
- Streamlit: Multi-turn conversation flows
- Response validation: Every response matches required JSON schema; all URLs verified from catalog; turn count ≤ 8 ; response time < 30 seconds

Conversation patterns tested:

- Vague first turn → Asks clarifying questions
 - Tech keyword → Recommends immediately
 - Mid-conversation changes → Updates list
 - Off-topic input → Refuses
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AI TOOLS USED

Claude (Anthropic): Used for code writing, debugging, and architecture design. I reviewed all code, tested locally, and modified for understanding. Code reflects actual implementation.

DEPLOYMENT

Deployed on Render: <https://shl-bot.onrender.com>

Live endpoints:

- GET /health → `{"status": "ok"}`
- POST /chat → Real recommendations from catalog

Both tested and working. GitHub: <https://github.com/srivathsav07/SHL-bot>